SHUBHANKAR SAHAI

Website-https://sahai-s.github.io// ◇ shubhankar.sahai.72@gmail.com

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EMPLOYMENT

LMSI, Mumbai *January-June 2027*
Visiting Post-doc

EDUCATION

University of California, San Diego *September 2021-November 2026 (expected)*
Thesis: *Derived algebraic geometry and Prismatic F-gauges*
PhD Student, Mathematics Advisor: Kiran Kedlaya

University of Michigan, Ann Arbor *January 2020 - December 2020*
Master of Science, Mathematics

University of Michigan, Ann Arbor *September 2014 - December 2017*
Bachelor of Science, Honors Mathematics and Interdisciplinary Physics High Honors

OTHER POSITIONS

Institute for Advanced Study, Princeton *September 2023-December 2023 and March 2024-April 2024*
Visiting Graduate Student, School of Mathematics Sponsor: Kiran Kedlaya

RESEARCH INTERESTS

Arithmetic and algebraic geometry, homotopy theory, representation theory

RECOGNITION

James B. Ax Fellowship, UCSD *September, 2021*

PAPERS/PREPRINTS

Duality of differential operators and algebraic de Rham cohomology *2024*
with C. Ji, C. Kothari, O. Li, S. Makarova and S. Venkatesh
Preprint available at https://arxiv.org/abs/2409.14322
Proves a Serre duality result for differential operators. Applied to terms of the E_1- Hodge-de Rham spectral sequence for smooth proper morphisms, gives another proof of relative Poincaré duality for relative de Rham cohomology.

Derived algebras on formal stacks and prismatic gauges. *2026*
Available https://arxiv.org/abs/2602.08942.
Develops some derived formal geometry to study the theory of derived algebras in quasi-coherent sheaves on derived formal stacks. Ends with Simpson style classification results for derived algebras on stacks modeling the Nygaard filtered prismatization of perfectoid rings. This has applications to the paper 'Prismatic F-gauges and Frobenius liftable schemes'.

De Rham affineness of the Nygaard filtered prismatization in positive characteristic. *2025*
In revision at IMRN. Preprint available at https://arxiv.org/abs/2512.19348.
Names a stronger affineness property (than Toën's affineness) exhibited by the Nygaard filtered prismatization

in positive characteristic as *de Rham affineness*. Shows that the Nygaard filtered prismatization in positive characteristic satisfies it.

Tannakian reconstruction in derived algebraic geometry. *2026*
Submitted at Forum of Mathematics, Sigma. Preprint available at `https://arxiv.org/abs/2608.00999`.
with J. Mundinger
We explain a derived refinement of Tannakian reconstruction in the sense of Lurie. The main insight (due to Nuiten-Toën) is that maps between animated rings can be recovered from maps between their module categories *and* the datum of induced functors compatible with a certain monad acting on the module categories.

Cotangent-étale $\mathbf{F}_p$-algebras and Cartier smoothness. *2026*
Draft available on request.
with R. Ishizuka
Cartier smoothness is a notion of regularity for non-Noetherians $\mathbf{F}_p$-algebras. For example regular rings over $\mathbf{F}_p$-are Cartier smooth. In this paper we give counter-examples of non-Noetherian rings which are Cartier smooth but don't have a flat Frobenius. This records the failure of Kunz's theorem beyond the Noetherian case.

Prismatic F-gauges and Frobenius liftable schemes. *2026*
In preparation. Poster at `https://sahai-s.github.io/Flift_poster.pdf`.
We show that the Nygaard filtered prismatization of a Frobenius liftable lci scheme is pulled back in a precise sense from the Hodge filtered de Rham stack of its lift. As a consequence we can study the syntomification of such an F-liftable schemes in terms of big Fontaine-Laffaille modules. Some open questions about constructing isomorphisms of ring stacks parametrising certain p-adic cohomology theories are discussed.

TALKS/ POSTER PRESENTATION

p-adic motives and Frobenius liftable schemes *December 4th, 2026*
WAGS 5-minute talks, University of Utah

Tannakian reconstruction in derived algebraic geometry *November 7, 2026*
Northeast Conference on Categorical Methods, Harvard CMSA

p-adic motives and Frobenius liftable schemes *November 4, 2026*
Purdue algebraic geometry seminar

p-adic motives and Frobenius liftable schemes *August 11, 2026*
Poster presentation at The Australian direction, University of Utah.

p-adic motives and Frobenius liftable schemes *May 22, 2026*
UCSD Algebraic Geometry seminar.

Algebraic de Rham cohomology and differential operators. *May 7, 2026*
UCSD Junior Algebraic Geometry seminar.

CONFERENCES ATTENDED/ WILL ATTEND

Northeast Conference on Categorical Methods *November 6-7, 2026*
CMSA, Harvard

Motivic Homotopy Theory: Connections and Applications *October 19-October 23, 2026*
SLMath, Berkeley

Australian direction *August 10-August 14, 2026*
University of Utah, Salt Lake City, UT, USA

2025 Summer Research Institute in Algebraic Geometry *July 14-August 1, 2025*
Colorado State University in Fort Collins, CO, USA

Arithmetic, K-theory and algebraic cycles *May 27-31, 2025*
Ohio State University, Ohio.

Informal seminar on p-adic geometry *September 16-20, 2024*
University of Chicago, Chicago.

Local Systems in Algebraic Geometry *May 7-10, 2024*
Ohio State University, Ohio.

Special Year on p-adic Arithmetic Geometry, Spring Workshop *11-15 March 2024*
IAS, Princeton

Special Year on p-adic Arithmetic Geometry, Fall Workshop *13-17 November 2023*
IAS, Princeton

Stacks Project Workshop 2023 *7-11 August 2023*
Ann Arbor, Michigan Mentor: Aise Johan de Jong

IHÉS summer school on algebraic K-theory *9-21 July 2023*
IHÉS, Bures-sur-Yvette

Algebraic geometry and cohomology in mixed characteristic *15-19 May 2023*
Northwestern University, Evanston

UNDERGRAD RESEARCH

Composition tableaux basis for Schur functors and the Plücker algebra *2020*
Preprint available at `https://arxiv.org/abs/1811.10687`

TEACHING

At UCSD

TA for Math 18 (Linear Algebra) *Fall 2026*

TA for Math 20-C (Calculus and Analytic geometry for Science and Engineering) *Summer Session 1 2026*

TA for Math 20-D (Differential equations) *Fall 2022*

TA for Math 20-E (Vector calculus) *Fall 2021, Winter 2022, Spring 2022*

TA for Math 109 (Introduction to mathematical thinking) *Winter 2025*

TA for Math 154 (Discrete mathematics and graph theory) *Winter 2026*

TA for Math 184 (Enumerative Combinatorics) *Spring 2026*

TA for Math 187-A (Cryptography) *Winter 2023, Spring 2023, Winter 2024, Spring 2024, Fall 2024, Spring 2025, Fall 2025*

SERVICE

At UCSD

Math Graduate Student Council (MGSC) mentor *2025-2026*
Mentoring an early grad student at UCSD.

Math Graduate Student Council (MGSC) mentorship chair *2024-2025*
Ran the mentorship program for the Graduate Student Council. This included pairing first year grad students with senior grad students and other relevant organizational duties.

RTG reading course in Local Class Field Theory *Spring 2023*
Students mentored- Jonathan Cummins, Xun Gong, Fuxiang Yang.

RTG reading course in commutative and homomlogical algebra. *Winter-Spring 2025*
Students mentored- Yu Sheng.
A somewhat intense reading course in commutative and homological algebra culminating in Huneke-Lyubeznik's proof of existence of big Cohen-Macaulay algebras.

Founder and organiser for Student Arithmetic Geometry seminar and Discord server *Summer 2022-2024*
The aim is to provide some structure and help to graduate students in arithmetic geometry at UCSD. Among other things, provides a structures space for beginning students to organize learning seminars, etc.

At Michigan

Minicourse on Algebraic K-theory *June 24-June 28, 2019*
Summer graduate student mini-course series at Michigan.

SEMINARS (CO) ORGANISED

Alterations *Spring 2026*
UCSD jointly organised with Ramanujan Srihari

Perverse sheaves (aka BBDG) seminar *Summer 2025*
UCSD/Boston University jointly organised with Matt Broe

Perverse sheaves in commutative algebra reading group *Winter and Spring 2025*
UCSD (I gave all the talks).

étale K-theory seminar *Fall 2024*
UCSD

Derived algebraic geometry *Spring 2023*
UCSD

Faltings' theorem learning seminar *Winter 2023*
UCSD jointly organised with Nathan Wenger.

Weil II learning seminar *Fall 2022*
UCSD

Abelian varieties learning seminar *Summer 2022*
UCSD

Prismatic cohomology reading group *Winter 2022*
UCSD

Algebraic stacks seminar *Winter 2022*
UCSD, jointly organised with Jack J. Garzella.

Crystalline cohomology reading group *Fall 2021*
UCSD

OTHER WORK EXPERIENCE.

Lab assistant *February 2018-December 2018*
The McMahon Cosmology Lab, University of Michigan, Ann Arbor[1]
This job involved writing technical code for machinery which constructs telescope lenses.

[1] Now at the University of Chicago.

Math tutor *February 2019-May 2019*
Math Lab, Department of mathematics, University of Michigan, Ann Arbor.

REFERENCES FOR POSTDOC APPLICATIONS

Advisor

Kiran Kedlaya *kedlaya@ucsd.edu*
UCSD

Research

Bhargav Bhatt *bhargav.bhatt@gmail.com*
IAS/Princeton

Aise Johan de Jong *dejong@math.columbia.edu*
Columbia University

Teaching

Glenn Tesler *gtesler@ucsd.edu*
UCSD